

VYAZ'MA, Russian Federation

WMO: 266950

Lat: 55.1570N

Lon: 34.3840E

Elev: 251

StdP: 98.35

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -20.6	(d) -26.3	(e) 0.4	(f) -23.6	(g) -23.0	(h) 0.5	(i) -20.2	(j) 9.6	(k) -1.9	(l) 8.4	(m) -2.3	(n) 1.0	(o) 350	(p) 0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 28.5	(d) 19.7	(e) 26.6	(f) 18.9	(g) 24.9	(h) 18.1	(i) 20.9	(j) 26.8	(k) 19.9	(l) 25.0	(m) 18.9	(n) 23.6	(o) 2.7	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.0	23.3	18.0	13.3	22.5	17.1	12.6	21.2	61.3	26.8	57.9	25.1	54.7	23.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.3	-28.7	30.4	3.4	2.4	-31.1	32.2	-33.1	33.6	-35.0	34.9	-37.4	36.6		
			-28.8	22.0	3.3	1.3	-31.2	22.9	-33.2	23.7	-35.0	24.4	-37.4	25.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-7.2	-7.0	-1.8	6.1	12.1	15.9	18.4	16.4	10.9	5.3	-0.6	-5.1		
	DBStd	10.29	6.40	6.78	4.80	4.82	4.76	3.56	3.15	3.43	3.70	4.14	5.22	6.34		
	HDD10.0	2517	534	474	366	136	32	1	0	0	32	154	319	467		
	HDD18.3	4822	793	708	624	367	199	91	37	82	223	404	569	726		
	CDD10.0	822	0	0	0	19	98	177	262	197	60	8	0	0		
	CDD18.3	85	0	0	0	0	7	17	41	20	1	0	0	0		
	CDH23.3	768	0	0	0	3	67	138	349	201	10	0	0	0		
	CDH26.7	181	0	0	0	0	9	19	91	61	1	0	0	0		
(14) Wind	WSAvg	2.8	3.1	3.0	3.1	2.8	2.7	2.5	2.1	2.3	2.3	2.9	3.1	3.2		
(15) Precipitation	PrecAvg	656	42	35	35	34	67	73	84	81	53	59	49	43		
	PrecMax	938	72	68	74	83	147	174	191	256	125	136	101	94		
	PrecMin	435	14	13	11	6	23	13	9	7	10	5	1	16		
	PrecStd	135	16	14	16	17	31	38	52	60	24	34	21	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	3.8	13.4	22.6	27.9	28.7	32.1	32.4	24.7	19.1	11.1	7.8		
		MCWB	3.7	2.4	7.5	12.2	18.5	19.4	20.9	20.5	17.4	13.0	9.6	7.4		
	2%	DB	2.3	2.6	9.1	19.7	25.3	26.8	28.9	27.9	21.8	15.0	8.5	5.2		
		MCWB	1.6	1.4	4.4	11.2	17.1	19.0	20.0	19.6	15.6	12.1	7.4	4.6		
	5%	DB	1.2	1.7	6.4	17.1	22.8	24.8	26.9	25.2	19.2	12.8	7.1	3.1		
		MCWB	0.6	0.7	3.1	10.4	15.5	17.8	19.4	18.4	14.4	10.7	6.1	2.5		
	10%	DB	0.3	1.0	4.2	14.6	20.4	22.8	25.0	23.1	17.1	11.0	5.7	1.6		
		MCWB	-0.2	0.2	1.9	8.9	13.9	16.9	18.7	17.4	13.4	9.3	4.9	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.9	2.9	7.7	13.3	19.9	21.1	22.5	21.6	18.6	14.3	10.0	7.5		
		MCDB	4.1	3.5	12.8	20.1	26.4	26.5	28.4	27.9	22.3	16.8	10.9	7.9		
	2%	WB	1.7	1.7	5.1	11.8	17.7	19.7	21.2	20.4	16.3	12.5	7.7	4.6		
		MCDB	2.1	2.2	8.1	18.1	23.6	25.0	27.2	26.4	20.1	14.6	8.3	5.1		
	5%	WB	0.7	0.9	3.5	10.7	16.4	18.6	20.2	19.3	15.0	10.9	6.2	2.6		
		MCDB	1.1	1.4	5.7	15.9	21.4	23.6	25.2	24.3	18.2	12.3	6.9	3.0		
	10%	WB	-0.1	0.3	2.2	9.5	15.0	17.5	19.2	18.1	13.7	9.6	5.0	1.2		
		MCDB	0.3	0.9	4.0	14.2	19.1	21.7	23.8	22.0	16.7	11.0	5.6	1.5		
(35) Mean Daily Temperature Range	MDBR		4.8	6.4	7.4	9.3	10.2	9.4	9.8	9.7	8.9	6.2	4.2	4.2		
	5% DB	MCDBR	3.3	4.3	10.2	13.2	13.6	12.4	13.1	13.4	12.7	9.1	5.4	3.8		
		MCWBR	3.4	3.7	6.4	6.5	6.4	5.8	5.4	6.0	7.0	6.0	4.6	3.7		
	5% WB	MCDBR	3.4	3.7	8.6	11.6	12.3	11.0	11.3	12.1	10.8	7.6	4.7	3.7		
		MCWBR	3.5	3.5	5.7	6.1	6.1	5.8	5.4	5.9	6.3	5.6	4.3	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.318	0.339	0.404	0.390	0.371	0.404	0.418	0.368	0.345	0.318	0.282		
	taud		2.129	2.078	2.137	2.165	2.312	2.387	2.311	2.262	2.375	2.387	2.248	2.131		
	Ebn at Noon		624	732	818	813	853	875	834	793	789	716	581	529		
	Edh at Noon		63	96	114	130	119	111	118	117	91	72	57	50		
(44) All-Sky Solar Radiation	RadAvg		0.64	1.53	3.04	4.04	5.28	5.59	5.40	4.35	2.88	1.34	0.52	0.38		
	RadStd		0.14	0.20	0.31	0.34	0.41	0.51	0.54	0.49	0.34	0.24	0.10	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.71	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.67	N/A	N/A	N/A	N/A	N/A	-129	-201	+93	N/A

Nomenclature: See separate page